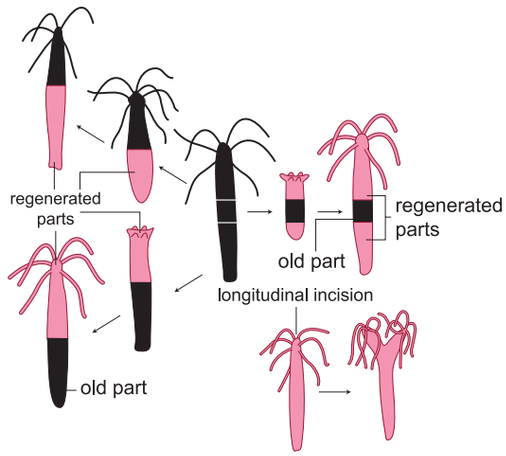


Fig. 1.11 Regeneration in *Hydra*

Regeneration is regrowth in the injured region. Regeneration was first studied in *Hydra* by **Abraham Trembley** in 1740. Regeneration is of two types, **morphallaxis** and **epimorphosis**. In morphallaxis the whole body grows from a small fragment e.g. *Hydra* and *Planaria*. When *Hydra* is accidentally cut into several pieces, each piece can regenerate the lost parts and develop into a whole new individual (**Fig. 1.11**). The parts usually retain their original polarity, with oral ends, by developing tentacles and aboral ends, by producing basal discs. **Epimorphosis** (**Fig. 1.12**) is the replacement of lost body parts. It is of two types, namely **reparative** and **restorative** regeneration. In reparative regeneration, only certain damaged tissue can be regenerated, e.g. human beings (liver tissue) whereas in restorative regeneration severed body parts can develop. e.g. star fish, tail of wall lizard.



Fig. 1.12 Restorative regeneration in star fish

1.3. Sexual reproduction

Sexual reproduction involves the fusion of male and female gametes to form a diploid zygote, which develops into a new organism. It leads to genetic variation. The types of sexual reproduction seen in animals are syngamy (fertilization) and conjugation. In **syngamy,** the fusion of two haploid gametes takes place to produce a diploid zygote. Depending upon the place where the fertilization takes place, it is of two types. In **external fertilization,** the fusion of male and female gametes takes place outside the body of female organisms in the water medium. e.g. sponges, fishes and amphibians. In **internal fertilization,** the fusion of male and female gametes takes place within the body of female organisms. e.g. reptiles, aves and mammals.



POWER OF REGENERATION

Sponge when macerated and squeezed through fine silk cloth, the cluster of cells pass through, and these can regenerate new sponges. This technique is used for cultivation of sponges.

Different kinds of syngamy (fertilization) are prevalent among living organisms.

a) Autogamy - The male and female gametes are produced by the same cell or same organism and both the gametes fuse together to form a zygote. e.g. *Actinosphaerium* and *Paramecium*.

b) Exogamy - The male and female gametes are produced by different parents and they fuse to form a zygote. So it is biparental. e.g. Human beings – dioecious or unisexual animal.

c) Hologamy - Lower organisms, sometimes the entire mature organisms do not form gametes but they themselves behave as gametes and the fusion of such mature individuals is known as **hologamy** e.g. *Trichonympha.*

XII Std Biology-Zoology Chapter-1 EM.indd 6 23/12/2022 10:57:06